

E-Commerce Business Analytics SQL Project

Author : Syamkumar Kummari

Skills Used

- SQL
- MySQL
- CTEs (Common Table Expressions)
- Window Functions
- Joins
- Aggregate Functions
- Subqueries
- Business Analytics
- Customer Segmentation
- Revenue Analysis
- Profit Analysis
- Churn Analysis
- Pareto Analysis
- Trend Analysis

Project Overview

This project focuses on analyzing E-Commerce business data using SQL to solve real-world business problems related to:

- Customer Analytics
- Product Performance
- Revenue & Profit Analysis
- Customer Segmentation
- Churn Analysis
- Payment Analysis
- Inventory Analysis
- Business Growth Trends

Key Highlights :

- Solved 80+ Business SQL Problems
- Implemented Advanced SQL Queries
- Used Window Functions & CTEs
- Performed Revenue & Profit Analysis
- Built Customer Segmentation Analysis
- Created Churn-Risk Customer Analysis
- Applied Pareto Analysis (80/20 Rule)
- Generated Business Insights from Data

11.Find top 10 customers based on total spending.

```
select c.customer_name, o.customer_id,sum(oi.final_price) as total_spending
from orders o
join customers c on
o.customer_id = c.customer_id
join order_items oi on
o.order_id = oi.order_id
where o.order_status in ('Shipped','Delivered')
group by customer_name ,customer_id
order by total_spending desc
limit 10;
```

	customer_name	customer_id	total_spending
▶	Abram Bhatia	1626	293662.59
	Yakshit Bail	2626	286991.89
	Ojas Joshi	841	286786.83
	Eva Dugal	4724	283904.51
	Sana Kibe	3352	264372.16
	Aarush Rajan	1268	260792.58
	Saira Basak	3285	259972.46
	Abram Bhat	2227	255338.95
	Himmat Chopra	487	254919.26
	Tara Biswas	2546	254387.57

Insights:

A small group of high-value customers contributes a significant portion of total revenue. These customers can be targeted through loyalty programs, premium offers, and personalized marketing campaigns.

25.Find category-wise profit.

```
select p.category , sum(oi.final_price - (p.cost_price * oi.quantity)) as profit from products p join order_items oi on
p.product_id = oi.product_id
join orders o on
oi.order_id = o.order_id
where o.order_status in ('Shipped','Delivered')
group by category;
```

	category	profit
▶	Books	8368439.45
	Home	8941511.15
	Beauty	10350934.23
	Sports	9033279.90
	Fashion	9700200.90
	Electronics	8027789.83

Insight:

Beauty and Fashion categories generate the highest profits, indicating strong customer demand and better profit margins. Electronics shows comparatively lower profitability despite high sales potential.

32.Find monthly revenue trend.

```
select year(o.order_date) as year_, monthname(o.order_date) as month_,sum(oi.final_price) as total_revenue from
orders o
```

```
join order_items oi on
```

```
o.order_id = oi.order_id
```

```
where o.order_status in ('Shipped','Delivered')
```

```
group by year_,month_,month(o.order_date)
```

```
order by year_,month(order_date),month_;
```

	year_	month_	total_revenue
	2023	May	3753803.82
	2026	May	4721991.47
	2024	March	6662499.22
	2025	September	6745539.37
	2025	October	6816241.39
	2024	December	6818678.96
	2025	August	6914949.55
	2024	February	6993796.71
	2024	July	7111465.70
	2025	June	7206014.10
	2024	April	7262043.34
	2024	November	7405069.80
	2026	February	7405105.43
	2025	July	7556272.70
	2026	April	7783972.78
	2023	June	7812213.26
	2024	June	7819876.11
	2025	February	7873884.46
	2026	January	7877222.71
	2024	October	7958327.89
	2023	September	7991213.55
►	2023	October	8034344.95
	2023	December	8103371.85
	2024	January	8134833.82
	2025	April	8322649.77

Insight:

Revenue shows steady growth across multiple months, with noticeable sales peaks during certain periods. This indicates seasonal purchasing behavior and business growth over time.

34.Find top profitable products.

```
select p.product_id , p.product_name ,sum(oi.final_price -(p.cost_price * oi.quantity)) as total_profit from orders o
```

```
join order_items oi on
```

```
o.order_id = oi.order_id
```

```
join products p on
```

```
p.product_id = oi.product_id
```

```
where o.order_status in ('Shipped','Delivered')
```

```
group by p.product_id , p.product_name
```

order by total_profit desc

limit 10;

	product_id	product_name	total_profit
▶	1574	Sapiente Product	154640.95
	392	Et Product	153044.78
	1754	Doloremque Product	152227.69
	1404	Saepe Product	152204.82
	981	Consectetur Product	149877.41
	1039	Veritatis Product	148876.27
	1538	Fugiat Product	147575.28
	1355	Natus Product	146446.92
	135	Recusandae Product	146296.57
	1822	Ea Product	143733.47

Insight:

Revenue shows steady growth across multiple months, with noticeable sales peaks during certain periods. This indicates seasonal purchasing behavior and business growth over time.

40. Find year-over-year growth.

with cte as (select year(o.order_date) as year_, sum(final_price) as total_revenue from orders o

join order_items oi on

o.order_id = oi.order_id

where o.order_status in ('Shipped','Delivered')

group by year_)

select year_, lag(total_revenue) over(order by year_) as previous_year_revenue,

round(((total_revenue - lag(total_revenue) over(order by year_)) / lag(total_revenue) over(order by year_)) * 100,2) as
YOY_growth from cte

order by year_;

	year_	previous_year_revenue	YOY_growth
▶	2023	NULL	NULL
	2024	61202376.54	51.65
	2025	92810604.75	2.14
	2026	94794827.79	-61.87

Insight:

The business experienced strong revenue growth in 2024 and moderate growth in 2025. However, 2026 shows a decline trend, indicating the need for improved sales and customer retention strategies.

41. Find top performing cities by sales.

select c.city , sum(oi.final_price) as total_sales from customers c

join orders o on

c.customer_id = o.customer_id

join order_items oi on

```
o.order_id = oi.order_id
```

```
where o.order_status in ('Shipped','Delivered')
```

```
group by c.city
```

```
order by total_Sales desc;
```

	city	total_sales
►	Hyderabad	43954762.08
	Bangalore	41595245.94
	Vijayawada	41256207.33
	Mumbai	40049113.85
	Kochi	39931689.51
	Chennai	39240638.02
	Delhi	38921336.14

Insight:

Hyderabad and Bangalore contribute the highest sales revenue, making them key regional markets for the business. These cities can be prioritized for expansion and targeted campaigns.

47. Find payment method-wise revenue.

```
select o.payment_method , sum(oi.final_price) as total_revenue from orders o
```

```
join order_items oi on
```

```
o.order_id = oi.order_id
```

```
where o.order_status in ('Shipped','Delivered')
```

```
group by o.payment_method
```

```
order by total_revenue desc;
```

	payment_method	total_revenue
►	Cash on Delivery	59838944.47
	Net Banking	57899847.00
	Debit Card	56194813.88
	UPI	55548944.73
	Credit Card	55466442.79

Insight:

Cash on Delivery generates the highest revenue contribution, showing strong customer preference for flexible payment options. Digital payment methods also contribute significantly to overall sales.

61. Rank products based on revenue generated.

```
select p.product_id , p.product_name , sum(oi.final_price) as revenue ,
```

```
rank() over (order by sum(oi.final_price) desc) as ranks
```

```
from products p
```

```
join order_items oi on
```

```

p.product_id = oi.product_id

join orders o on

o.order_id = oi.order_id

where o.order_status in('Shipped','Delivered')

group by p.product_id,p.product_name;

```

	product_id	product_name	revenue	rank
▶	661	Facere Product	499249.32	1
	1754	Doloremque Product	473659.60	2
	1355	Natus Product	464571.54	3
	392	Et Product	462108.73	4
	1496	Eligendi Product	454897.42	5
	1538	Fugiat Product	452503.06	6
	1574	Sapiente Product	450870.03	7
	1635	Nemo Product	448297.11	8
	1039	Veritatis Product	441705.82	9
	1404	Saepe Product	440204.47	10
	981	Consectetur Product	428992.55	11
	396	Laborum Product	426668.88	12
	826	Tenetur Product	424011.16	13
	135	Recusandae Product	422578.78	14
	1387	Fugit Product	421347.81	15
	1477	Amet Product	418391.59	16
	385	Dicta Product	417276.53	17
	1093	Ilum Product	415776.73	18
	812	Esse Product	411780.86	19
	1061	Atque Product	408787.00	20
	1269	Occaecati Product	407441.67	21
	1044	Accusamus Product	401712.32	22
	1370	Asperiores Product	399603.30	23
	159	Quo Product	392984.00	24
	808	Quibusdam Product	389612.77	25

Insight:

The highest-ranked products generate substantial revenue compared to lower-ranked products, highlighting the importance of identifying and promoting top-performing inventory items.

63.Find running total of monthly revenue.

```

with cte as ( select year(o.order_date) as year_, month(o.order_date) as month_num,monthname(o.order_date) as
month_,sum(oi.final_price) as total_revenue from orders o

```

```

join order_items oi on

```

```

o.order_id = oi.order_id

```

```

where o.order_status in ('Shipped','Delivered')

```

```

group by year_,month_,month(o.order_date)

```

```

order by year_,month(order_date),month_

```

```

select year_,month_,total_revenue,sum(total_revenue) over (order by year_,month_num) as running_total from cte;

```

	year_	month_	total_revenue	running_total
►	2023	May	3753803.82	3753803.82
	2023	June	7812213.26	11566017.08
	2023	July	8759380.42	20325397.50
	2023	August	8358394.04	28683791.54
	2023	September	7991213.55	36675005.09
	2023	October	8034344.95	44709350.04
	2023	November	8389654.65	53099004.69
	2023	December	8103371.85	61202376.54
	2024	January	8134833.82	69337210.36
	2024	February	6993796.71	76331007.07
	2024	March	6662499.22	82993506.29
	2024	April	7262043.34	90255549.63
	2024	May	9178437.81	99433987.44
	2024	June	7819876.11	107253863.55
	2024	July	7111465.70	114365329.25
	2024	August	8984454.54	123349783.79
	2024	September	8481120.85	131830904.64
	2024	October	7958327.89	139789232.53
	2024	November	7405069.80	147194302.33
	2024	December	6818678.96	154012981.29
	2025	January	8346475.59	162359456.88
	2025	February	7873884.46	170233341.34
	2025	March	9338985.59	179572326.93
	2025	April	8322649.77	187894976.70
	2025	May	8323634.66	196218611.36

Insight:

The cumulative revenue trend shows continuous business growth over time. Running totals help measure long-term revenue performance and monitor overall business progress.

64.Find month-over-month sales growth.

```

with cte as (select year(o.order_date) as year_,
                    month(o.order_date) as month_num,
                    monthname(o.order_date) as month_,
                    sum(oi.final_price) as revenue ,
                    lag(sum(oi.final_price)) over(order by year(o.order_date) , month(o.order_date)) as previous
                from orders o
                join order_items oi on
                o.order_id = oi.order_id
                where o.order_status in ('Shipped','Delivered')
                group by year_,month_num,month_)
select year_,month_,revenue,previous , round(((revenue - previous) / previous)*100,2) as growth from cte;

```

	year_	month_	revenue	previous	growth
►	2023	May	3753803.82	NULL	NULL
	2023	June	7812213.26	3753803.82	108.11
	2023	July	8759380.42	7812213.26	12.12
	2023	August	8358394.04	8759380.42	-4.58
	2023	September	7991213.55	8358394.04	-4.39
	2023	October	8034344.95	7991213.55	0.54
	2023	November	8389654.65	8034344.95	4.42
	2023	December	8103371.85	8389654.65	-3.41
	2024	January	8134833.82	8103371.85	0.39
	2024	February	6993796.71	8134833.82	-14.03
	2024	March	6662499.22	6993796.71	-4.74
	2024	April	7262043.34	6662499.22	9.00
	2024	May	9178437.81	7262043.34	26.39
	2024	June	7819876.11	9178437.81	-14.80
	2024	July	7111465.70	7819876.11	-9.06
	2024	August	8984454.54	7111465.70	26.34
	2024	September	8481120.85	8984454.54	-5.60
	2024	October	7958327.89	8481120.85	-6.16
	2024	November	7405069.80	7958327.89	-6.95
	2024	December	6818678.96	7405069.80	-7.92
	2025	January	8346475.59	6818678.96	22.41
	2025	February	7873884.46	8346475.59	-5.66
	2025	March	9338985.59	7873884.46	18.61
	2025	April	8322649.77	9338985.59	-10.88
	2025	May	8323634.66	8322649.77	0.01

Insight:

Sales growth fluctuates across months, indicating changing customer demand and seasonal business patterns. Positive growth periods can help identify successful sales strategies and peak demand months.

66.Find products contributing to 80% revenue (Pareto Analysis).

```

with cte as (select p.product_name , sum(oi.final_price) as revenue from products p
join order_items oi on
p.product_id = oi.product_id
join orders o on
o.order_id = oi.order_id
where o.order_status in ('Shipped','Delivered')
group by p.product_name),
pareto as ( select product_name , revenue , sum(revenue) over(order by revenue desc) as running_total,
sum(revenue) over() as total_revenue,
round((sum(revenue) over(order by revenue desc) /sum(revenue) over())*100,2) as percentage from cte)
select * from pareto
where percentage <= 80;

```


	product_name	revenue	running_total	total_revenue	percentage
►	Eligendi Product	3094983.67	3094983.67	284948992.87	1.09
	Optio Product	2790485.65	5885469.32	284948992.87	2.07
	Earum Product	2760282.01	8645751.33	284948992.87	3.03
	Qui Product	2709700.10	11355451.43	284948992.87	3.99
	Vel Product	2576393.56	13931844.99	284948992.87	4.89
	Fugiat Product	2563419.22	16495264.21	284948992.87	5.79
	Libero Product	2537829.55	19033093.76	284948992.87	6.68
	Corporis Product	2521025.75	21554119.51	284948992.87	7.56
	Consequatur Product	2493408.92	24047528.43	284948992.87	8.44
	Atque Product	2429663.78	26477192.21	284948992.87	9.29
	Esse Product	2420226.34	28897418.55	284948992.87	10.14
	At Product	2416217.90	31313636.45	284948992.87	10.99
	Repellendus Product	2390867.96	33704504.41	284948992.87	11.83
	Fugit Product	2335649.76	36040154.17	284948992.87	12.65
	Voluptate Product	2312336.45	38352490.62	284948992.87	13.46
	Soluta Product	2293901.25	40646391.87	284948992.87	14.26
	Culpa Product	2281245.91	42927637.78	284948992.87	15.07
	Harum Product	2246436.88	45174074.66	284948992.87	15.85
	Pariatur Product	2240610.79	47414685.45	284948992.87	16.64
	Dolorum Product	2224642.76	49639328.21	284948992.87	17.42
	Laborum Product	2211617.62	51850945.83	284948992.87	18.20
	Neque Product	2207277.09	54058222.92	284948992.87	18.97
	Quibusdam Product	2190637.03	56248859.95	284948992.87	19.74
	Quasi Product	2174983.08	58423843.03	284948992.87	20.50
	Accusantium Product	2170956.63	60594799.66	284948992.87	21.27

Insight:

A relatively small percentage of products contributes to the majority of total revenue, following the Pareto Principle (80/20 rule). Focusing on these products can maximize business performance.

67.Segment customers into:High Value,Medium Value,Low Value

with cte as (select c.customer_id, c.customer_name , sum(oi.final_price) as spending from customers c

join orders o on

o.customer_id = c.customer_id

join order_items oi on

o.order_id = oi.order_id

where o.order_status in ('Shipped','Delivered')

group by c.customer_id,c.customer_name

order by spending desc)

select customer_id , customer_name , spending ,

case

when spending <= 150000 then 'Low Value'

when spending <= 250000 then 'Medium value'

when spending > 250000 then 'High value'

end as segmentation

from cte;

	customer_id	customer_name	spending	segmentation
▶	1626	Abram Bhatia	293662.59	High value
	2626	Yakshit Bail	286991.89	High value
	841	Ojas Joshi	286786.83	High value
	4724	Eva Dugal	283904.51	High value
	3352	Sana Kibe	264372.16	High value
	1268	Aarush Rajan	260792.58	High value
	3285	Saira Basak	259972.46	High value
	2227	Abram Bhat	255338.95	High value
	487	Himmat Chopra	254919.26	High value
	2546	Tara Biswas	254387.57	High value
	2127	Arhaan Bhatt	247861.30	Medium value
	3214	Vanya Edwin	245135.15	Medium value
	2889	Tiya Dâ€™Alia	241444.35	Medium value
	4498	Stuvan Wagle	240363.54	Medium value
	1305	Onkar Bava	239619.84	Medium value
	2307	Raunak Sule	233688.41	Medium value
	3625	Hunar Rangan...	230956.10	Medium value
	4917	Ryan Rangana...	227432.35	Medium value
	3708	Siya Sha	226269.77	Medium value
	1088	Yashvi Sinha	225284.46	Medium value
	2044	Zara Sangha	223010.90	Medium value
	2351	Ritvik Datta	222484.08	Medium value
	2026	Kavya Kala	216975.65	Medium value
	1189	Nehmat Vora	216685.22	Medium value
	1330	Heer Bala	215182.03	Medium value

Insight:

Customers are segmented into High, Medium, and Low Value groups based on spending behavior. High-value customers contribute significantly to revenue and should be prioritized through retention and loyalty strategies.

70.Find churn-risk customers.

with cte as (select c.customer_id,c.customer_name , max(o.order_date) as last_order_date from customers c

join orders o on

c.customer_id = o.customer_id

group by c.customer_id ,c.customer_name)

select * from cte

where last_order_date < current_date() - interval 6 month;

	customer_id	customer_name	last_order_date
▶	1	Divij Viswanathan	2025-08-09
	2	Ishita Balasubramanian	2025-04-16
	7	Azad Sanit	2024-09-06
	9	Anvi Thakur	2025-03-14
	10	Umang Sethi	2025-05-15
	11	Anya Chakraborty	2024-12-06
	12	Trisha Krishna	2025-06-18
	15	Hridaan Sahni	2025-05-15
	16	Tejas Goswami	2025-07-15
	17	Amani Manne	2025-08-13
	19	Emir Dube	2025-04-04
	23	Aarush Bhat	2025-07-31
	28	Tarini Bath	2023-07-21
	29	Kashvi Guha	2025-10-20
	30	Aradhya Rana	2023-05-22
	38	Dishani Sandal	2025-09-09
	39	Nirvi Master	2025-11-22
	42	Mehul Sawhney	2025-03-09
	43	Jayan Kale	2023-11-15
	44	Ira Bhalla	2025-10-10
	46	Charvi Chaudhari	2025-11-03
	47	Himmat Subramaniam	2025-01-26
	48	Oorja Arora	2025-09-19
	51	Anvi Din	2024-03-16
	53	Charvi Dey	2025-07-06

Insight:

Customers who have not placed orders in recent months are at higher risk of churn. Re-engagement campaigns and personalized offers can help improve customer retention.

74.Are discounts increasing or reducing profits?

```
select case
```

```
    when oi.discount_percent <=10 then 'Low Discount'
```

```
        when oi.discount_percent <=20 then 'Medium Discount'
```

```
        when oi.discount_percent >20 then 'High Discount'
```

```
end as discount_category,
```

```
sum(oi.final_price - (p.cost_price*oi.quantity)) as profit
```

```
from products p
```

```
join order_items oi on
```

```
p.product_id = oi.product_id
```

```
join orders o on
```

```
o.order_id = oi.order_id
```

```
where o.order_status in ('Shipped','Delivered')
```

```
group by discount_category
```

```
order by profit desc;
```

	discount_category	profit
▶	Low Discount	29243114.08
	Medium Discount	18077962.41
	High Discount	7101078.97

Insight:

Lower discount levels generate significantly higher profits compared to high-discount sales. Excessive discounts may reduce overall profitability despite increasing sales volume.

76.Which categories are growing fastest?

```
with cte as (select p.category, year(o.order_date) as year_, sum(oi.final_price) as current_year from orders o
join order_items oi on
o.order_id = oi.order_id
join products p on
p.product_id = oi.product_id
where o.order_status in ('Shipped','Delivered')
group by p.category,year_
order by category),
previous as ( select category ,year_,current_year ,lag(current_year) over(PARTITION BY category order by year_) as
previous_year from cte)
select category ,year_,current_year,previous_year, round((((current_year - previous_year) / previous_year)*100,2) as growth
from previous
order by growth desc;
```

	category	year_	current_year	previous_year	growth
►	Books	2024	15477377.24	9247573.86	67.37
	Electronics	2024	14079169.68	8627527.58	63.19
	Home	2024	15030553.36	9719358.44	54.65
	Beauty	2024	16704390.97	11537116.80	44.79
	Sports	2024	15141367.06	10485151.74	44.41
	Fashion	2024	16377746.44	11585648.12	41.36
	Fashion	2025	17347538.03	16377746.44	5.92
	Sports	2025	16007139.57	15141367.06	5.72
	Beauty	2025	17658468.40	16704390.97	5.71
	Home	2025	15746832.08	15030553.36	4.77
	Books	2025	14833061.73	15477377.24	-4.16
	Electronics	2025	13201787.98	14079169.68	-6.23
	Beauty	2026	7014050.65	17658468.40	-60.28
	Electronics	2026	5239313.51	13201787.98	-60.31
	Sports	2026	6306837.39	16007139.57	-60.60
	Home	2026	6090262.52	15746832.08	-61.32
	Fashion	2026	6198529.25	17347538.03	-64.27
	Books	2026	5292190.47	14833061.73	-64.32
	Beauty	2023	11537116.80	NULL	NULL
	Books	2023	9247573.86	NULL	NULL
	Electronics	2023	8627527.58	NULL	NULL
	Fashion	2023	11585648.12	NULL	NULL
	Home	2023	9719358.44	NULL	NULL
	Sports	2023	10485151.74	NULL	NULL

Insight:

Certain categories show strong yearly growth trends, indicating changing customer preferences and emerging market demand. These categories can be prioritized for future investment and expansion.

80.Which customers are likely VIP customers?

```
with cte as (select c.customer_id, c.customer_name , sum(oi.final_price) as spending from customers c
join orders o on
```

```

o.customer_id = c.customer_id

join order_items oi on

o.order_id = oi.order_id

where o.order_status in ('Shipped','Delivered')

group by c.customer_id,c.customer_name

order by spending desc),

segmentation as (select customer_id , customer_name , spending ,

    case

        when spending <= 150000 then 'Low Value'

        when spending <= 250000 then 'Medium value'

        when spending > 250000 then 'High value'

    end as segmentation

    from cte)

select customer_name ,spending from segmentation

where segmentation = 'High value';

```

	customer_name	spending
▶	Abram Bhatia	293662.59
	Yakshit Bail	286991.89
	Ojas Joshi	286786.83
	Eva Dugal	283904.51
	Sana Kibe	264372.16
	Aarush Rajan	260792.58
	Saira Basak	259972.46
	Abram Bhat	255338.95
	Himmat Chopra	254919.26
	Tara Biswas	254387.57

Insight:

High-spending customers represent the business's VIP customer base. Retaining these customers through loyalty benefits and personalized engagement can significantly improve long-term revenue.